Application of Machine Learning Methods to Predict Building Damage Levels in the Nepal Earthquake

Machine Learning Mini-Project Report

Gandhar Gokhale PES1UG24CS167

Chetana Iyer PES1UG24CS130

1. Abstract

This project investigates supervised machine learning methods for predicting building damage grades from building characteristics recorded following the 2015 Nepal earthquake. The dataset contains 260,601 labelled buildings and 86,868 unlabelled test buildings. Exploratory analysis followed by stratified train-validation split and a Random Forest baseline. On the held-out validation set, the baseline achieved 73.19% accuracy, a macro F1-score of 0.6801, and a weighted F1-score of 0.7277. A sampled comparison between Random Forest and Extra Trees and an aggregated feature-importance analysis were also performed. Finally, predictions were generated for the provided test records.

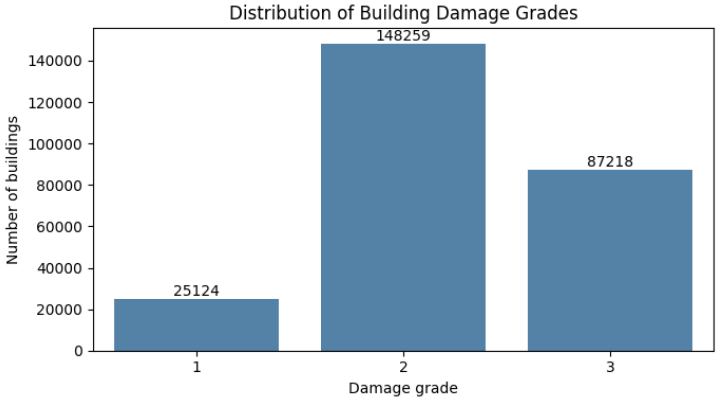
2. Problem Statement and Objectives

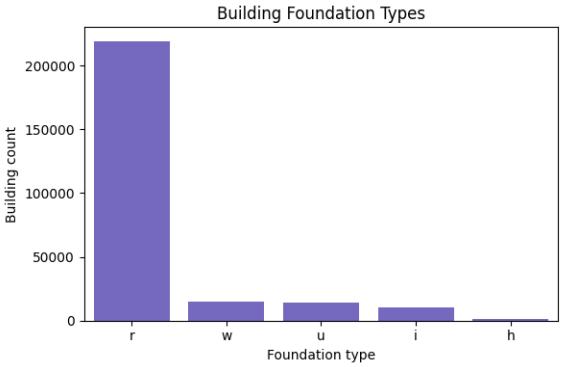
The objective is to classify each building into one of three damage grades (1, 2, or 3) using recorded geographic, structural, and building-use characteristics. The project aims to (i) inspe ct and validate the data, (ii) train and evaluate a baseline classifier, (iii) compare alternative tree-ensemble models under consistent sampled conditions, (iv) identify features used heavily by the baseline model, and (v) produce predictions in the format expected by the supplied submission template.

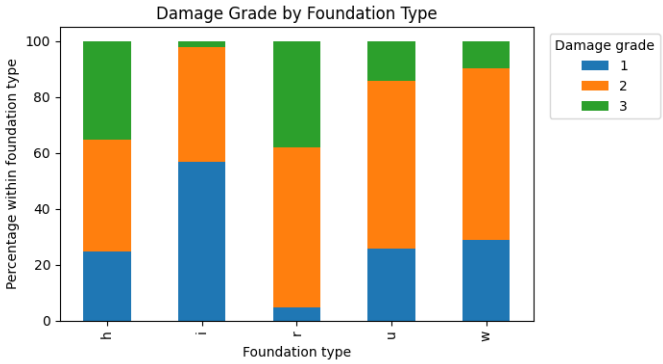
3. Dataset and Exploratory Data Analysis

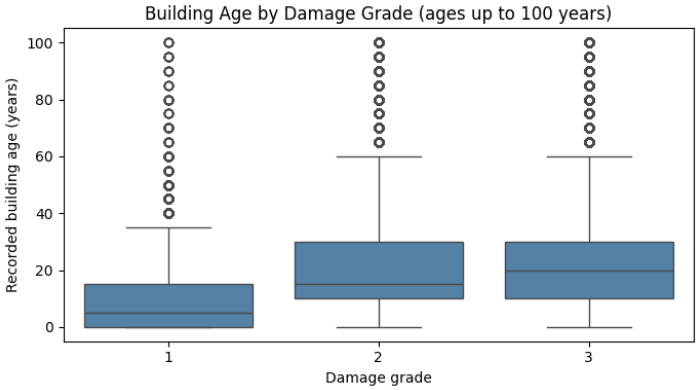
The input consists of train_values.csv (260,601 rows and 39 columns), train_labels.csv (260,601 rows and 2 columns), test_values.csv (86,868 rows and 39 columns), and submission_format.csv (86,868 rows and 2 columns). The training features and labels were checked for unique building IDs and matched by building_id using a one-to-one merge. The feature and label files contained no missing values. There were 16,191 duplicate feature rows when building_id was excluded; these were retained because different buildings can share identical recorded characteristics.

Damage grades were imbalanced: Grade 1 accounted for 25,124 buildings (9.64%), Grade 2 for 148,259 (56.89%), and Grade 3 for 87,218 (33.47%). The EDA also examined data types , foundation-type frequencies, damage proportions by foundation type, and building-age distributions. A small number of recorded ages were unusually high, including 1,390 records with age 995; these values were retained rather than removed without evidence about their meaning.









4. Methodology

The building_id column was used to align records and create the output file, but was excluded from model features. The target was damage_grade. A stratified 80:20 split produced 208,480 training records and 52,121 validation records. Text categorical features and geographic identifiers (geo_level_1_id, geo_level_2_id, and geo_le vel_3_id) were one-hot encoded within a scikit-learn pipeline; remaining numeric and binary features were passed through. The encoder was fitted on training data and configured to ignore previously unseen categories during validation or prediction.

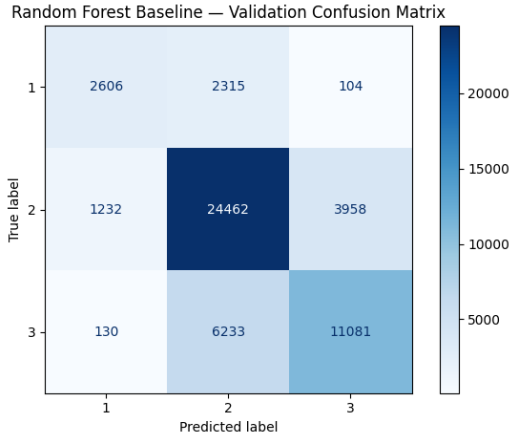
The baseline was a RandomForestClassifier with 100 trees, random_state=42, and n_jobs=-1. Model quality was measured using accuracy, macro F1, weighted F1, per-class precision/recall/F1, and a confusion matrix. Macro F1 was emphasized because the target classes are imbalanced. For a computationally manageable model comparison, Random Forest (50 trees, max_depth=30, min_samples_leaf=2) and Extra Trees (100 trees, max_depth=30, min_samples_leaf=2) were fitted on the same 52,120-row sample and evaluated on the same validation set.

5. Results

Model	Training rows	Accuracy	Macro F1
Random Forest baseline (full split)	208,480	0.7319	0.6801
Random Forest (sample)	52,120	0.6532	0.4687
Extra Trees (sample)	52,120	0.6479	0.4548

The full-split Random Forest baseline achieved 73.19% accuracy, 0.6801 macro F1, and 0.7277 weighted F1. Its class wise F1-scores were approximately 0.58 for Grade 1, 0.78 for Grade 2, and 0.68 for Grade 3. confusion matrix shows that Grade 1 was often predicted as Grade 2, while many Grade 3 examples were also misclassified as Grade 2. This shows that distinguishing adjacent damage grades remains a challenge.

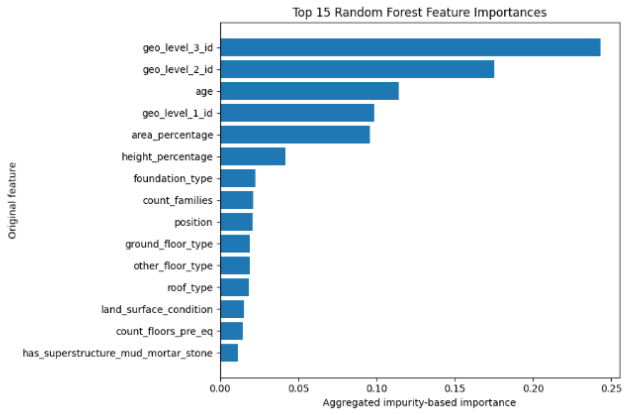
The sampled Random Forest slightly outperformed sampled Extra Trees on accuracy and ma cro F1. These two sampled models are directly comparable because they used the same training sample and validation set. Their results should not be compared as if they had the same training size as the full-split baseline.



6. Feature Importance

The baseline Random Forest's impurity-based feature importances were aggregated from one-hot encoded columns back to their original input features. The five highest values were geo_level_3_id (0.2430), geo_level_2_id (0.1754), age (0.1144), geo_level_1_id (0.0983), and area_percentage (0.0957). Height percentage, foundation type, family count, position, and floor/roof characteristics followed.

These results indicate which features the fitted model relies on for its splits; th ey do not demonstrate that the features cause building damage. Impurity-based importance can also favour high-cardinality features, so the geographic feature values should be interpreted cautiously.



7. Final Test Predictions

The fitted baseline Random Forest was reused to predict damage_grade for the 86,868 records in test_values.csv. Predictions were joined to the test building IDs and saved to /kaggle/working/nepal_earthquake_submission.csv. The generated output has 86,868 rows and two columns (building_id and damage_grade), matching the expected submission structure. The predicted counts were 6,545 Grade 1, 55,3 32 Grade 2, and 24,991 Grade 3. The supplied test set has no local ground-truth labels, so test accuracy cannot be computed from the available files. Note that this submission uses the baseline trained on the 80% training split; the optional retraining on all labelled data was interrupted.

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Submission saved successfully!
File: /kaggle/working/nepal_earthquake_submission.csv
Shape: (86868, 2)

Predicted damage-grade distribution:
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	count
damage_grade	
1	6545
2	55332
3	24991

	building_id	damage_grade
0	300051	2
1	99355	2
2	890251	2
3	745817	1
4	421793	3

8. Conclusion and Limitations

The project demonstrates a complete baseline workflow for predicting Nepal earthquake building damage grades: data validation and label alignment, exploratory analysis, preprocessing, model training, validation, feature-importance analysis, and test prediction export. The baseline Random Forest achieved 73.19% validation accuracy and 0.6801 macro F1. Its performance was strongest for the most frequent class, Grade 2, and weaker for Grade 1. The sampled comparison found slightly better performance for Random Forest than E xtra Trees under the tested settings.

Limitations include class imbalance, unusual recorded ages, the use of impurity-based feature importance, and the absence of ground-truth test labels for local evaluation. The final prediction file was produced using the fitted baseline model rather than a model retrained on all labelled data. Future work could test class weighting, alternative classifiers, carefully controlled hyperparameter tuning, permutation importance, and evaluation on an independently labelled test set. This model is an educational analysis tool and must not replace structural engineering inspection or official damage assessment.